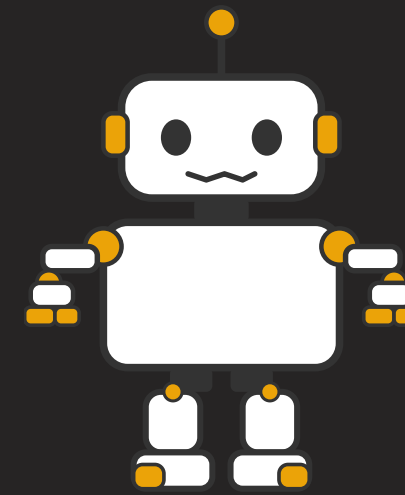




Coding Challenge

Robot Work Allocation System

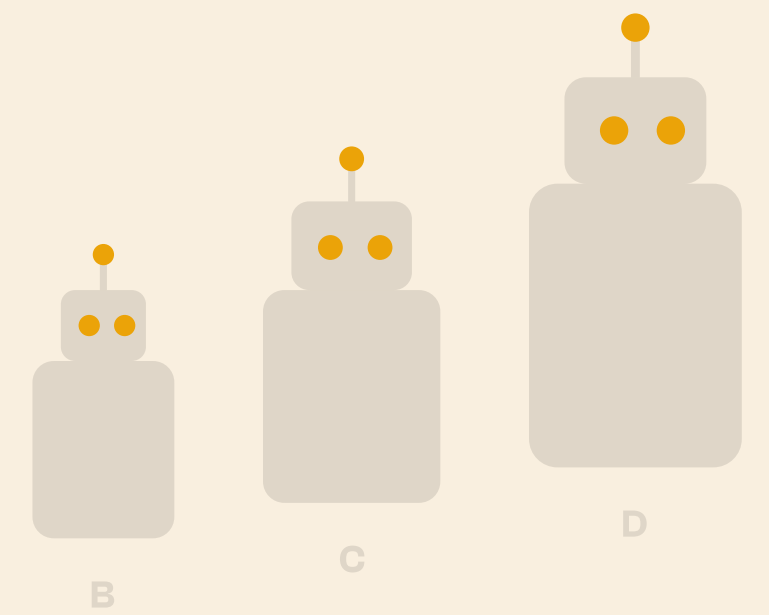
Terminal-Based Application for EverBot Solutions

COMPANY OVERVIEW

EverBot Solutions

A robotics services company that uses specialized robots for performing client's automated work.

- * Robots are categorised into **three types**: Bravo, Charlie, Delta
- * Each robot works **once per day** for its max hours, then recharges
- * Clients submit work requests measured in **total number of hours**
- * Your job is to build a system that assigns robots to fulfil client work **as efficiently as possible**, based on different strategies in each level.



Robot Types

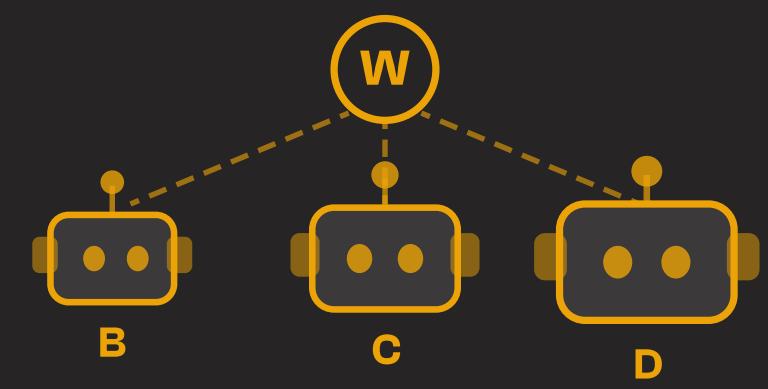
ROBOT TYPE	WORKING HOURS / DAY	CHARGING COST / DAY
 Bravo	3 hours	\$2
 Charlie	5 hours	\$3
 Delta	8 hours	\$4

General Rules

- * Robots **cannot** be used more than once per day
- * The robots' combined working hours need not match the exact working hours of the client's work. E.g: if client's working hours = 16, robots combined working hours can be 17 or 18, but not less than 16.
- * If work cannot be fulfilled, show a **clear error message**

Error: Insufficient robot capacity to complete the requested work.

LEVEL 1



Robot Category Distribution

Evaluate performance across robot categories for future purchases

LEVEL 1

Category Distribution Strategy

GOALS

- * **Primary:** Include multiple categories when feasible
- * Respect robot availability constraints
- * Robot hours provided $\geq$ hours requested
- * Minimise excess hours

EXAMPLES

- * 17 hrs needed → pick **Bravo** (extra)
- * 24 hrs needed → pick **Delta** (extra)
- * 21 hrs needed → pick **Charlie** (extra)

Pick the robot that minimises excess hours

LEVEL 1

Example: 16 Hours Requested

INPUT

```
Enter number of robots available:
Bravo: 2
Charlie: 3
Delta: 2

Enter client work hours:
16
```

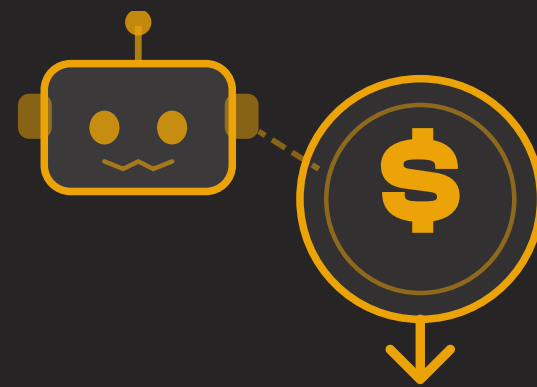
OUTPUT

```
Robot Assignment

Bravo: 1
Charlie: 1
Delta: 1

Total Work Hours Provided: 16
Client Work Hours Requested: 16
```

LEVEL 2



Cost Optimised Allocation

Maximise profits by minimising robot charging costs

Cost Efficiency Strategy

- * Allocate robots purely based on cost efficiency
- * Ignore the Level 1 requirement of using multiple robot types
- * Minimise **Total Charging Cost** while fulfilling requested hours

EXAMPLE 1:

INPUT

```
Enter number of robots available:  
Bravo: 2  
Charlie: 3  
Delta: 2  
  
Enter client work hours:  
20
```

OUTPUT

```
Cost Optimized Allocation  
  
Charlie: 1  
Delta: 2  
  
Total Hours Provided: 21  
Total Charging Cost: $11
```

EXAMPLE 2:

INPUT

```
Enter number of robots available:  
Bravo: 2  
Charlie: 2  
Delta: 3  
  
Enter client work hours:  
6
```

OUTPUT

```
Cost Optimized Allocation  
  
Bravo: 2  
  
Total Hours Provided: 6  
Total Charging Cost: $4
```

ADDITIONAL REQUIREMENT

Level 1 vs Level 2 Comparison

The system must also compare the Level 1 solution vs Level 2 solution

For Example 1:

Level 1 Cost: \$12

Level 2 Cost: \$11

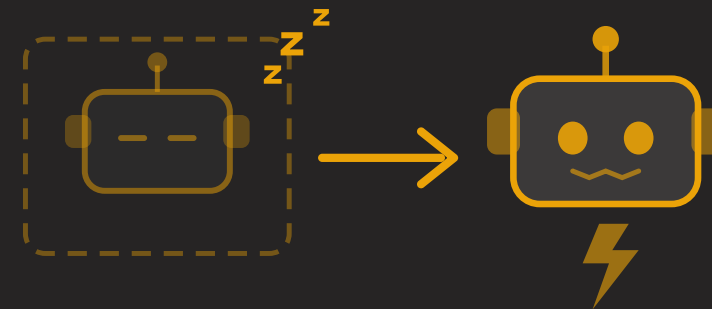
Cost Difference: \$1

Insight:

Level 1 strategy resulted in \$1 additional cost due to mandatory usage of multiple robot categories.

This helps the company **understand the financial impact** of performance analysis strategy

LEVEL 3



Standby Robot Activation

Activate warehouse standby robots when active capacity is insufficient

LEVEL 3

Standby Activation Strategy

The company also maintains a **warehouse of standby robots** that are not currently active.

- * Activate when client hours **exceed active capacity**
- * Determine **which** standby robots and **how many** of each type
- * Same **working hours & charging cost** rules apply
- * Choose the **cost-optimised** standby option

EXAMPLE 1:

```
Active robots:
Bravo: 1
Charlie: 1
Delta: 1

Client work request:
21 hours

Maximum Active Capacity:
3 + 5 + 8 = 16 hours
```

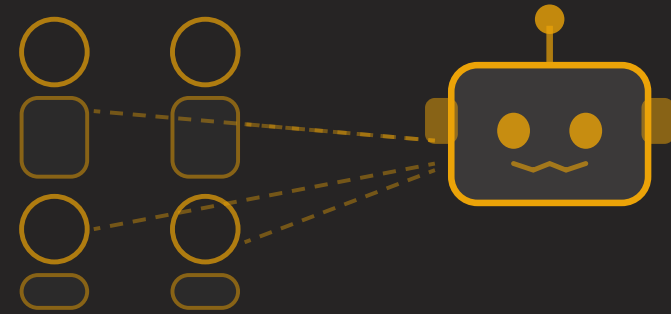
EXPECTED OUTPUT

```
Active Robot Capacity: 16 hours
Client Work Requested: 21 hours

Additional Standby Robots Required:
Bravo: 2 - cost $4
or
Delta: 1 - cost $4
or
Charlie: 1 - cost $3

The output should show Charlie: 1
as it's the cost optimised one
```

LEVEL 4



Serving Multiple Clients

Scale up to handle multiple concurrent client work requests

LEVEL 4

Multi-Client Allocation

Allow robots for multiple clients based on the following conditions:

- * Prioritise allocation based on **highest hours requested**
- * If active robots are insufficient, list **standby robots** needed to satisfy clients' working hours
- * The application should pick the number of clients based on the input value for client working hours such as comma or space separated values, or single value

```
Client working hours: 20
```

```
Client working hours: 12,16,17,10,21
```

```
Client working hours: 12 16 17 10 21
```

Error Handling

IMPOSSIBLE ALLOCATION

```
Error: Unable to allocate  
at least one robot from  
each category with the  
available inventory.
```

ZERO ROBOTS

```
Error: No robots  
available for assignment.
```

INVALID INPUT

```
Error: Work hours must be  
a positive integer.
```

Additional Constraints

1

Robot counts must be
non-negative integers

2

Client work hours must be
positive integers

3

Each robot can only be used
once per allocation

BONUS

Optional Advanced Features

MULTIPLE CLIENTS

Allow processing multiple work requests

```
Clients: [16, 10, 22, 7]
```

ALLOCATION SUMMARY

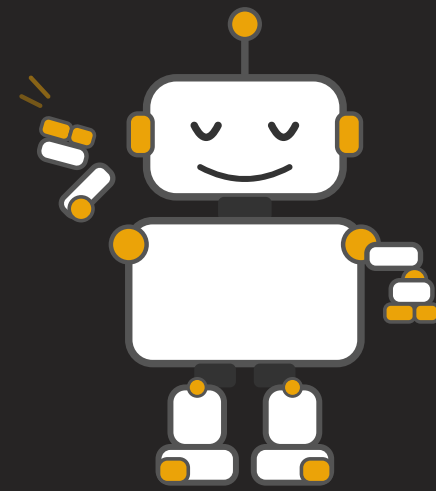
Display summary after all clients

```
Total Robots Used  
Total Charging Cost  
Avg Robot Utilization
```

EFFICIENCY METRICS

Display the following:

```
Bravo utilization  
Charlie utilization  
Delta utilization
```



Thank you!
